

Secure Private-Network AI

4 layers

Network
isolation

VPC with private
subnets for all
compute

Security Groups + NACLs
for least-privilege
network access

VPC Endpoints
(PrivateLink) for Bedrock,
SageMaker, S3, KMS — no
public internet

Identity &
access

IAM execution roles per
workload (not shared)

SCPs at OU level to
prevent service-level
bypasses

Attribute-Based Access
Control (ABAC)

Data
governance

Amazon Macie for PII
detection in training
data

KMS CMK encryption for
model artifacts, S3,
SageMaker volumes

Lake Formation column-
level security on feature
data

Audit &
compliance

CloudTrail for all
Bedrock/SageMaker API
calls

SageMaker Model Cards
for model documentation
and governance

Config Rules for
continuous compliance
checking

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